
AWS Cloud Services Cheat Sheet

Compute Services

- **EC2:** Virtual servers, resizable compute, pay-as-you-go, AMI, instance types
- **Lambda:** Serverless, event-driven, auto scaling, pay per request
- **Lightsail:** Simple VPS, preconfigured apps, fixed pricing
- **Elastic Beanstalk:** PaaS, automatic deployment, scaling, monitoring
- **ECS:** Container orchestration, Docker support, integrates with Fargate
- **EKS:** Managed Kubernetes, scalable containers, integrates with AWS
- **Fargate:** Serverless containers, no server management, works with ECS/EKS
- **Batch:** Batch processing jobs, optimized compute, scheduling
- **Outposts:** AWS on-premises, hybrid cloud, consistent infra
- **Wavelength:** Edge computing, low latency, 5G integration
- **Local Zones:** Extend AWS closer to users, low-latency apps

Storage Services

- **S3:** Object storage, unlimited capacity, versioning, lifecycle
- **EBS:** Block storage, EC2 volumes, persistent, snapshots
- **EFS:** File storage, shared, scalable, NFS protocol
- **FSx:** Managed file systems, Windows/Linux, high performance
- **S3 Glacier:** Archival storage, low cost, long-term backup
- **AWS Backup:** Centralized backup, automation, cross-region
- **Storage Gateway:** Hybrid storage, file/block/tape, on-prem integration
- **Snow Family:** Data migration, edge computing, Snowcone/Snowball/Snowmobile

Database Services

- **RDS:** Managed relational DB, MySQL/PostgreSQL/Oracle/SQL Server, backups
- **Aurora:** High-performance, MySQL/PostgreSQL compatible, auto scaling
- **DynamoDB:** NoSQL, key-value store, millisecond latency, serverless
- **Redshift:** Data warehousing, analytics, scalable clusters

- **ElastiCache:** In-memory caching, Redis/Memcached, improves performance
- **Neptune:** Graph DB, relationships, social/network data
- **DocumentDB:** MongoDB compatible, document-based, scalable
- **QLDB:** Ledger DB, immutable, cryptographically verifiable
- **Keyspaces:** Managed Cassandra, scalable, NoSQL
- **DMS:** Database migration, heterogeneous/homogeneous, minimal downtime

Networking & CDN

- **VPC:** Isolated network, subnets, route tables, security groups
- **ELB:** Distribute traffic, auto scaling, high availability
- **Route 53:** DNS, domain registration, health checks, routing
- **Direct Connect:** Dedicated network, low latency, high bandwidth
- **Transit Gateway:** Connect multiple VPCs, hub-and-spoke, simplified routing
- **Global Accelerator:** Improves global application performance
- **CloudFront:** CDN, low latency, edge locations
- **PrivateLink:** Secure private connectivity, no public internet
- **VPN:** Encrypted connections, site-to-site, client VPN
- **App Mesh:** Service mesh, microservices communication, observability

Security, Identity & Compliance

- **IAM:** Access control, users/groups/roles, least privilege
- **SSO:** Centralized sign-in, multiple accounts, SAML
- **KMS:** Encryption keys, FIPS compliance, integrated with services
- **Shield:** DDoS protection, Standard & Advanced
- **WAF:** Web application firewall, blocks SQL injection/XSS
- **Secrets Manager:** Securely stores & rotates API keys/passwords
- **Certificate Manager:** SSL/TLS certs, auto-renewal, free public certs
- **CloudHSM:** Hardware security module, FIPS 140-2 Level 3
- **Cognito:** User sign-up/sign-in, federated identities
- **Artifact:** Compliance reports, audit documentation

- **Inspector:** Vulnerability scanning, EC2/ECR, automated
- **GuardDuty:** Threat detection, ML-based monitoring
- **Security Hub:** Centralized security view, compliance checks
- **Detective:** Investigate security issues, visualize relationships
- **Audit Manager:** Automates evidence collection, compliance reporting

Analytics

- **Athena:** Serverless query, S3 data, SQL queries
- **EMR:** Hadoop/Spark clusters, big data processing, scalable
- **Glue:** ETL, data catalog, integrates with Athena & Redshift
- **Kinesis:** Real-time streaming, analytics, video ingestion
- **QuickSight:** BI tool, dashboards, ML insights
- **Data Pipeline:** Data workflow automation, scheduling
- **Lake Formation:** Build data lakes, centralized access control
- **OpenSearch:** Search & analytics, logs, monitoring
- **Redshift Spectrum:** Query S3 data, extends Redshift

Machine Learning / AI

- **SageMaker:** Build/train/deploy ML models, Jupyter, auto scaling
- **Rekognition:** Image/video analysis, face/object detection
- **Comprehend:** NLP, sentiment analysis, entity recognition
- **Polly:** Text-to-speech, lifelike voices, multiple languages
- **Lex:** Chatbots, voice/text, NLP engine
- **Translate:** Neural machine translation, real-time
- **Transcribe:** Speech-to-text, audio transcription
- **Textract:** Extract text/data from documents, OCR
- **Forecast:** Predictive forecasting, time-series ML
- **Personalize:** Recommendation engine, personalization

Management & Governance

- **CloudFormation:** Infrastructure as Code, templates, automation

- **CloudTrail:** API activity logs, auditing, compliance
- **CloudWatch:** Metrics, logs, alarms, dashboards
- **Config:** Resource inventory, compliance auditing, change tracking
- **Systems Manager:** Centralized management, automation, patching
- **Trusted Advisor:** Cost/security/performance checks
- **Organizations:** Multi-account management, consolidated billing
- **Control Tower:** Landing zone setup, governance, guardrails
- **Budgets:** Cost control, alerts, monthly tracking
- **Cost Explorer:** Visualize costs, usage analytics, forecasting

Application Integration

- **SQS:** Message queue, decoupling, FIFO/Standard queues
- **SNS:** Pub/Sub, notifications, fan-out
- **MQ:** Managed ActiveMQ/RabbitMQ, migration
- **Step Functions:** Workflow orchestration, serverless, state machines
- **EventBridge:** Event-driven apps, integrates with SaaS, automation

AWS Well-Architected Framework

Operational Excellence Automation, monitoring, continuous improvement

Security Identity, encryption, detection, compliance

Reliability Fault tolerance, recovery, scaling

Performance Efficiency Right sizing, monitoring, optimization

Cost Optimization Pay-as-you-go, eliminate waste, resource tagging

Sustainability Efficient resources, energy reduction, long-term impact

Pillar	Focus / Goal	Some Key Considerations
Operational Excellence	Running & monitoring systems, evolving procedures	Automate changes, respond to events, define standards & playbooks
Security	Protecting data, systems, and assets	Identity & access management, encryption, logging & detection
Reliability	Ensure workload can recover from failures	Fault injection, multi-AZ, graceful degradation, change management
Performance Efficiency	Use resources efficiently, adapt to demand	Right-sizing, selecting appropriate resource types, monitoring
Cost Optimization	Avoid unnecessary cost, scale economically	Pay per use, right-sizing, reserve vs on-demand, eliminate waste
Sustainability	Minimize environmental impact	Efficient utilization, resource sharing, energy-aware architectures

Support Plan	Key Keywords / Features
Basic	Free, 24/7 access to customer service, documentation, whitepapers, Trusted Advisor (limited)
Developer	Paid, business hours email support, <24h response for general guidance, <12h for system impaired, ideal for early development/testing
Business	Paid, 24/7 phone/email/chat support, <1h response for urgent issues, Trusted Advisor checks (full), Cloud Support API, AWS Support Concierge
Enterprise	Paid, 24/7 support, <15min critical issue response (urgent), Technical Account Manager (TAM), Infrastructure Event Management (IEM), Well-Architected Reviews, proactive guidance